Zhenyu Liao

Curriculum Vitae

May 2026

https://zhenyu-liao.github.io
zhenyu_liao@hust.edu.cn
Male, Chinese, born in 28/08/1992.

Education

2019	**Ph.D.**	Mathematics and Informatics	L2S, CentraleSupélec, University of Paris-Saclay, France.
2016	**M.Sc.**	Signal and Image Processing	University of Paris-Saclay, France.
2014	**B.E.**	Optical & Electronic Information	Huazhong university of Science and Technology, China.

Professional Experiences

- May-June, 2025: **CRM-Simons Visiting Scholar**, Centre de recherches mathématiques (CRM), Université de Montréal, as a part of the thematic program "Mathematical Foundations of Data Science." (*Unable to attend due to visa issue.*)
- June-July, 2024: **ANR-LabEx-CIMI Visiting Professor**, Centre International de Mathématiques et Informatique de Toulouse (CIMI), Toulouse, France.
- 2021-now: **Associate Professor** at School of Electronic Information and Communications, Huazhong University of Science and Technology (HUST).
- 2020-2021: **Postdoctoral Scholar** at ICSI and Department of Statistics, University of California, Berkeley, hosted by Michael Mahoney.

Honors and Awards

- 2025: IEEE Senior Member.
- 2025: Recipient of the MATRIX-Simons Travel Grant, MATRIX, Creswick, Australia.
- 2023: Recipient of Talent Program of Hubei Province, Hubei, China.
- 2021: Recipient of the Wuhan Youth Talent, Wuhan, China.
- 2021: Recipient of East Lake Youth Talent Program Fellowship of HUST, Wuhan, China.
- 2019: 2nd prize of ED STIC Doctoral Prize, University Paris-Saclay, France.
- 2016: Recipient of the Supélec Foundation Ph.D. Fellowship, France.

Publications

I've co-authored a book and more than 30 papers published on conferences and journals on machine learning, signal processing, and statistics, with more than 1 300 citations and h-index = 16 according to Google Scholar.

Books

1. Romain Couillet and **Zhenyu Liao**. *Random Matrix Methods for Machine Learning*. Cambridge University Press, 2022. DOI: 10.1017/9781009128490.

Papers in conference proceedings

1. Yue Xu and **Zhenyu Liao**. An Improved Convergence Analysis of Gossip Methods for Large Random Graphs. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. 2026, pp.22562–22566. DOI: 10.1109/ICASSP55912.2026.11461477.
2. Yessin Moahker, Malik Tiomoko, Cosme Louart, and **Zhenyu Liao**. A Random Matrix Perspective of Echo State Networks: From Precise Bias-Variance Characterization to Optimal Regularization. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. 2026, pp.20716–20720. DOI: 10.1109/ICASSP55912.2026.11464627.
3. Jiuqi Wei, **Zhenyu Liao**, Ruoyu Han, Quanqing Xu, Chuanhui Yang, and Themis Palpanas. TaCo: Data-adaptive and Query-aware Subspace Collision for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 4(3) (May 2026). DOI: 10.1145/3802118.
4. Chengmei Niu, **Zhenyu Liao**, Zenan Ling, and Michael W. Mahoney. Fundamental Bias in Inverting Random Sampling Matrices with Application to Sub-sampled Newton. In: *Proceedings of the 42nd International Conference on Machine Learning (**ICML**)*. Vol. 267. Proceedings of Machine Learning Research. PMLR, 2025, pp.46649–46692. https://proceedings.mlr.press/v267/niu25c.html.
5. Xiaoyi Mai and **Zhenyu Liao**. The Breakdown of Gaussian Universality in Classification of High-dimensional Mixtures. In: *International Conference on Learning Representations (**ICLR**)*. 2025. https://openreview.net/forum?id=UrKbn51HjA.

6. Jiuqi Wei, Xiaodong Lee, **Zhenyu Liao**, Themis Palpanas, and Botao Peng. Subspace Collision: An Efficient and Accurate Framework for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (**SIGMOD**)* **3**(1) (Feb. 2025). DOI: 10.1145/3709729.
7. Wei Yang, Zhengyu Wang, Xiaoyi Mai, Zenan Ling, Robert Caiming Qiu, and **Zhenyu Liao**. Inconsistency of ESPRIT DoA Estimation for Large Arrays and a Correction via RMT. In: *2024 32nd European Signal Processing Conference (**EUSIPCO**)*. Aug. 2024, pp.2722–2726. DOI: 10.23919/EUSIPCO63174.2024.10715081.
8. Zenan Ling, Longbo Li, Zhanbo Feng, Yixuan Zhang, Feng Zhou, Robert C. Qiu, and **Zhenyu Liao**. Deep Equilibrium Models Are Almost Equivalent to Not-so-deep Explicit Models for High-dimensional Gaussian Mixtures. In: *Proceedings of the 41st International Conference on Machine Learning (**ICML**)*. Vol. 235. PMLR, 2024, pp.30585–30609. https://proceedings.mlr.press/v235/ling24a.html.
9. Yi Song, Kai Wan, **Zhenyu Liao**, Hao Xu, Giuseppe Caire, and Shlomo Shamai. An Achievable and Analytic Solution to Information Bottleneck for Gaussian Mixtures. In: *2024 IEEE International Symposium on Information Theory (**ISIT**)*. July 2024, pp.2460–2465. DOI: 10.1109/ISIT57864.2024.10619077.
10. Yuanjie Wang, Zhanbo Feng, and **Zhenyu Liao**. FedRF-Adapt: Robust and Communication-Efficient Federated Domain Adaptation via Random Features. In: *2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (**ICASSPW**)*. Apr. 2024, pp.615–619. DOI: 10.1109/ICASSPW62465.2024.10626024.
11. Lingyu Gu, Yongqi Du, Yuan Zhang, Di Xie, Shiliang Pu, Robert Qiu, and **Zhenyu Liao**. "Lossless" Compression of Deep Neural Networks: A High-dimensional Neural Tangent Kernel Approach. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 35. Curran Associates, Inc., 2022, pp.3774–3787. https://proceedings.neurips.cc/paper_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html.
12. Hafiz Tiomoko Ali, **Zhenyu Liao**, and Romain Couillet. Random matrices in service of ML footprint: ternary random features with no performance loss. In: *International Conference on Learning Representations (**ICLR**)*. 2022. https://openreview.net/forum?id=qwULHx9zld.
13. **Zhenyu Liao** and Michael W Mahoney. Hessian Eigenspectra of More Realistic Nonlinear Models. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 34. Curran Associates, Inc., 2021, pp.20104–20117. https://papers.nips.cc/paper/2021/hash/a7d8ae4569120b5bec12e7b6e9648b86-Abstract.html.
14. Michal Derezinski, **Zhenyu Liao**, Edgar Dobriban, and Michael Mahoney. Sparse sketches with small inversion bias. In: *Proceedings of Thirty Fourth Conference on Learning Theory (**COLT**)*. Vol. 134. PMLR, 2021, pp.1467–1510. https://proceedings.mlr.press/v134/derezinski21a.html.
15. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. Sparse Quantized Spectral Clustering. In: *International Conference on Learning Representations (**ICLR**)*. 2021. https://openreview.net/forum?id=pBqLS-7KYAF.
16. Fanghui Liu, **Zhenyu Liao**, and Johan Suykens. Kernel Regression in High Dimension: Refined Analysis beyond Double Descent. In: *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics (**AISTATS**)*. Vol. 130. PMLR, 2021, pp.649–657. http://proceedings.mlr.press/v130/liu21b.html.
17. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. A Random Matrix Analysis of Random Fourier Features: Beyond the Gaussian Kernel, A Precise Phase Transition, and the Corresponding Double Descent. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 33. Curran Associates, Inc., 2020, pp.13939–13950. https://papers.nips.cc/paper/2020/hash/a03fa30821986dff10fc66647c84c9c3-Abstract.html.
18. Michal Derezinski, Feynman T Liang, **Zhenyu Liao**, and Michael W. Mahoney. Precise expressions for random projections: Low-rank approximation and randomized Newton. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 33. Curran Associates, Inc., 2020, pp.18272–18283. https://papers.nips.cc/paper/2020/hash/d40d35b3063c11244fbf38e9b55074be-Abstract.html.
19. **Zhenyu Liao** and Romain Couillet. On Inner-Product Kernels of High Dimensional Data. In: *2019 IEEE 8th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (**CAMSAP**)*. IEEE. 2019, pp.579–583. DOI: 10.1109/CAMSAP45676.2019.9022455.
20. Xiaoyi Mai, **Zhenyu Liao**, and Romain Couillet. A Large Scale Analysis of Logistic Regression: Asymptotic Performance and New Insights. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. IEEE. 2019, pp.3357–3361. DOI: 10.1109/ICASSP.2019.8683376.

21. Romain Couillet, **Zhenyu Liao**, and Xiaoyi Mai. Classification Asymptotics in the Random Matrix Regime. In: *The 26th European Signal Processing Conference (**EUSIPCO**)*. IEEE. 2018, pp.1875–1879. DOI: 10.23919/EUSIPCO.2018.8553034.
22. **Zhenyu Liao** and Romain Couillet. The Dynamics of Learning: A Random Matrix Approach. In: *Proceedings of the 35th International Conference on Machine Learning (**ICML**)*. Vol. 80. PMLR, 2018, pp.3072–3081. http://proceedings.mlr.press/v80/liao18b.html.
23. **Zhenyu Liao** and Romain Couillet. On the Spectrum of Random Features Maps of High Dimensional Data. In: *Proceedings of the 35th International Conference on Machine Learning (**ICML**)*. Vol. 80. PMLR, 2018, pp.3063–3071. http://proceedings.mlr.press/v80/liao18a.html.
24. **Zhenyu Liao** and Romain Couillet. Random Matrices Meet Machine Learning: A Large Dimensional Analysis of LS-SVM. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. IEEE. 2017, pp.2397–2401. DOI: 10.1109/ICASSP.2017.7952586.

Journal papers

1. **Zhenyu Liao** and Michael W. Mahoney. Random Matrix Theory for Deep Learning: Beyond Eigenvalues of Linear Models. *IEEE Signal Processing Magazine* **43**(2) (2026), 93–106. DOI: 10.1109/MSP.2025.3618012.
2. Zhengyu Wang, Wei Yang, Xiaoyi Mai, Zenan Ling, **Zhenyu Liao**, and Robert C. Qiu. A Large-dimensional Analysis of ESPRIT DoA Estimation: Inconsistency and a Correction via RMT. *IEEE Transactions on Signal Processing* (2026), 1–14. DOI: 10.1109/TSP.2026.3683673.
3. Zhanbo Feng, Yuanjie Wang, Jie Li, Fan Yang, Jiong Lou, Tiebin Mi, Robert. C. Qiu, and **Zhenyu Liao**. Robust and Communication-Efficient Federated Domain Adaptation via Random Features. *IEEE Transactions on Knowledge and Data Engineering* **37**(3) (2025), 1411–1424. DOI: 10.1109/TKDE.2024.3510296.
4. Jingcheng Wang, Shaoliang Zhang, Jianming Cai, **Zhenyu Liao**, Christian Arenz, and Ralf Betzholz. Robustness of random-control quantum-state tomography. *Phys. Rev. A* **108** (2 2023), 022408. DOI: 10.1103/PhysRevA.108.022408.
5. Yacine Chitour, **Zhenyu Liao**, and Romain Couillet. A geometric approach of gradient descent algorithms in linear neural networks. *Mathematical Control and Related Fields* **13**(3) (2023), 918–945. DOI: 10.3934/mcrf.2022021.
6. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. A random matrix analysis of random Fourier features: beyond the Gaussian kernel, a precise phase transition, and the corresponding double descent. *Journal of Statistical Mechanics: Theory and Experiment* **2021**(12) (2021), 124006. DOI: 10.1088/1742-5468/ac3a77.
7. **Zhenyu Liao** and Romain Couillet. A Large Dimensional Analysis of Least Squares Support Vector Machines. *IEEE Transactions on Signal Processing* **67**(4) (2019), 1065–1074. DOI: 10.1109/TSP.2018.2889954.
8. Cosme Louart, **Zhenyu Liao**, and Romain Couillet. A Random Matrix Approach to Neural Networks. *The Annals of Applied Probability* **28**(2) (2018), 1190–1248. DOI: 10.1214/17-AAP1328.

Research Grants

1. 2026-2029: **PI**, National Natural Science Foundation of China, general program "*Compression of Large-scale Transformer-based Models with Random Matrix Methods: From Scaling Law to Dynamic Adaptive Model Compression*" (NSFC-12571561), ¥440K, together with Cosme Louart at SDS, CUHK-SZ.
2. 2025–2030: Co-PI, National Key R&D Program of China (Youth Scientist Project), "*Foundations of Machine Learning for Understanding Large Models*" (2025YFA1018600), ¥1M/3M. **PI**: Jun Shu, School of Mathematics and Statistics, Xi'an Jiaotong University, and jointly conducted with Hongxin Wei and Zeng Li at SUSTech, as well as Zenan Ling at HUST.
3. 2026–2028: **PI**, HUST Basic Research Support Program (Mathematics Track), "*Theory and Methods for Large-scale Machine Learning via RMT*" (2025BRSXB0004), ¥380K.
4. 2024-2026: **PI**, Guangdong Key Lab of Mathematical Foundations for Artificial Intelligence Open Fund "*Generalization Theory for Transformer-based Models via Random Matrix Methods*" (OFA00003), ¥100K, with Jeff Yao as co-PI.
5. 2023-2025: **PI**, National Natural Science Foundation of China, youth program "*Fundamental Limits of Pruning Deep Neural Network Models via Random Matrix Methods*" (NSFC-62206101), ¥300K.
6. 2022-2024: **contributor**, National Natural Science Foundation of China, grants for "*Mathematical Foundations for Future Communications (Information Theory)*" (NSFC-12141107), ¥3M. PI: Robert C. Qiu.
7. 2021-2022: **PI**, CCF-Hikvision Open Fund, "*Random Matrix Theory and Information Bottleneck for Neural Network Compression*" (20210008), ¥280K, with Kai Wan as co-PI.

8. 2018-2021: **contributor**, NSF Research Grant, "*Combining Stochastics and Numerics for Improved Scalable Matrix Computations*" (NSF-1815054), $500K, PI: Michael W. Mahoney.
9. 2018-2021: **contributor**, Programme d'Investissements d'avenir, "*GSTATS IDEX DataScience Chair*", University of Grenoble-Alpes, €300K, PI: Romain Couillet.
10. 2014-2017, **contributor**, French National Research Agency, "*Random Matrix Theory for Large Dimensional Graphs*" (ANR-14-CE28-0006), €300K, PI: Romain Couillet.

Professional Services

Organization of Scientific Activities

1. **Area Chair** of ICML 2025–2026, NeurIPS 2025–2026, ICLR 2025–2026, AISTATS 2026, IJCNN 2025–2026.
2. **Associate Editor** of Springer Statistics and Computing (since 2025).
3. Co-organizer (together with Rishabh Dudeja, Junjie Ma, and Arian Maleki) the "Universality and Dynamics in High-Dimensional Learning and Inference" Workshop at 2026 IEEE International Symposium on Information Theory (ISIT 2026), June 2026, Guangzhou, China.
4. Co-organizer (together with Dirk Slock) of the Special Session on "Large-Dimensional Structures and Methods for Signal Processing, Communication, and Machine Learning" at 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026), May 2026, Barcelona, Spain.
5. Co-organizer (together with Pengkun Yang and Cong Fang) of 2025 Mathematical and Statistical Foundations of Foundation Models Workshop, July 2025, Changbaishan, China, see playback.
6. **General Co-chair** of 10th EAI International Conference on loT as a Service (EAI IoTaaS 2024), 2024.
7. Co-organizer (together with Jeff Yao, Shurong Zheng, Jiang Hu and Dandan Jiang) of 2025 Random Matrix Theory and Application (RMTA 2025) Conference, Jan 2025, Changchun, China, with more than 150 attendees, see playback.
8. Co-organizer of the 1st High-dimensional Learning Dynamics (HiLD) workshop at ICML 2023, together with Mihai Nica, Courtney Paquette, Elliot Paquette, Andrew Saxe, and René Vidal.
9. Co-organizer of the 2022 Joint workshop on "Math for Data Science" between HUST and University of Paris-Saclay, together with Yacine Chitour. The workshop had more than 40 000 online attendees, see playback.

Peer Reviewing

- **Research grants**:
  - Referee of European Research Council (ERC).
  - External reviewer of Natural Sciences and Engineering Research Council of Canada (NSERC).
  - Referee of National Natural Science Foundation of China (NSFC).
  - Reviewer of Israel Science Foundation (ISF).
- **Conferences**: ICLR, ICML, NeurIPS, SODA, AISTATS, AAAI, ECAI, IEEE-ICASSP, IEEE-CAMSAP.
- **Journals**:
  - Applied mathematics, statistical physics, statistics, and data science: SIAM Review (SIREV), Foundations of Computational Mathematics (FoCM), Physical Review X (PRX), Journal of Mathematical Physics, Springer Statistics and Computing (STCO), SIAM Journal on Scientific Computing (SISC), Random Matrices: Theory and Applications (RMTA), Latin American Journal of Probability and Mathematical Statistics (ALEA), Annals of Applied Probability (AAP), Electronic Communications in Probability (ECP) Electronic Journal of Statistics (EJS), Journal of Applied Mathematics and Computing (JAMC), AIMS Mathematics.
  - ML and AI: Journal of Machine Learning Research (JMLR), IEEE Trans. on Pattern Analysis and Machine Intelligence (IEEE-TPAMI), IEEE Signal Processing Letters (SPL), IEEE Trans. on Signal Processing (IEEE-TSP), IEEE Trans. on Multimedia (IEEE-TMM), Engineering Applications of Artificial Intelligence (EAAI), Frontiers of Computer Science, Neural Networks (NEUNET), IEEE Trans. on Neural Networks and Learning Systems (IEEE-TNNLS), IEEE Transactions on Industrial Informatics (IEEE-TII), Transactions on Machine Learning Research (TMLR), Pattern Recognition (PR), Neural Processing Letters (NPL).

Tutorials and Invited Talks

1. Invited talk on "A Random Matrix Approach to Neural Networks: From Linear to Nonlinear, and from Shallow to Deep" at "A day on Random Matrix Theory and Deep Learning" workshop, Rome Center on Mathematics for Modeling and Data ScienceS (RoMaDS), University of Rome Tor Vergata, Rome, September, 2025.

2. Invited talk on "Random matrix theory for deep learning: opportunities and challenges" at "Log-gases in Caeli Australi" research program, MATRIX institute of Creswick, Australia, August, 2025.
3. Invited talk on "Examples and Counterexamples of Gaussian Universality in High-dimensional Machine Learning" at "CSIAM-BDAI 2025" Conference, Guilin, China, July 2025.
4. Invited talk on "A Random Matrix Approach to Neural Networks: From Linear to Nonlinear, and from Shallow to Deep" at Gouxionghui, July 2025. See online playback.
5. Invited talk on "Inversion Bias in Random Matrices: Characterization and Implications for Randomized Numerical Linear Algebra" at "The 10th International Forum on Statistics," Beijing, China, July, 2025.
6. Invited talk on "Inversion Bias in Random Matrices: Characterization and Implications for Randomized Numerical Linear Algebra" at Workshop on "Random matrices and high-dimensional learning dynamics," Centre de recherches mathématiques (CRM), Canada, June, 2025.
7. Invited talk on "A Random Matrix Approach to Neural Networks: From Linear to Nonlinear, and from Shallow to Deep" at Fudan University Forum for Young Scholars and PhD Students in Data Science and Deep Learning, Shanghai, April 2025.
8. Invited talk on "Examples and Counterexamples of Gaussian Universality in Large-dimensional Machine Learning" at RMTA 2024 Conference, Changchun, China, Jan, 2025.
9. Invited talk on "Random Matrix Methods for Explicit and Implicit Neural Networks" at The 4th China Big Data Statistics Forum, Xuzhou, China, October, 2024.
10. 6 **H mini-course** on "Random Matrix Theory for Machine Learning", thematic trimester on "beyond classical regimes in statistical inference and machine learning," Centre International de Mathématiques et Informatique de Toulouse (CIMI), Toulouse, France, 2024.
11. Invited talk on "Recent Advances in Random Matrix Methods for Deep Learning Theory" at CSML 2024, Shanghai, China, August, 2024.
12. Invited talk on "A Random Matrix Approach to Explicit and Implicit Deep Neural Networks" at Institut de Mathématiques de Toulouse (IMT), Toulouse, France, July, 2024.
13. 12 **H mini-course** on "Random Matrix Theory for Modern Machine Learning: New Intuitions, Improved Methods, and Beyond" at Institut de Recherche en Informatique de Toulouse (IRIT), Toulouse, France, July, 2024.
14. 5 **H short course** on "Random Matrix Theory and Its Applications in ML" at Jiangsu Normal University, China, 2024. Links to recordings: part 1 on RMT and part 2 on its application to deep learning.
15. 3 **H mini-course** on "Random Matrix Theory in Deep Learning: An Introduction", Northeast Normal University, Changchun, Nov, 2023.
16. Invited talk on "Neural Tangent Kernel in High Dimensions: From Theory to Practice," Beijing Jiaotong University School of Computer and Information Technology, Beijing, Oct, 2023.
17. Invited talk on "Random Matrix Methods for Machine Learning: with Applications to "Lossless" Compression and Training of DNNs", International Chinese Statistical Association (ICSA) China Conference, Chengdu, July, 2023.
18. Invited talk on "On the inversion bias of randomized sketching", Shenzhen Conference on Random Matrix Theory and Applications (RMTA 2023), CUHK-Shenzhen, June, 2023.
19. Invited talk on "Random Matrix Methods for Machine Learning: with Applications to "Lossless" Compression and Training of DNNs", Guangzhou Institute of International Finance, Guangzhou, June, 2023.
20. Invited talk on "Random Matrix Methods for Machine Learning: with Applications to "Lossless" Compression and Training of DNNs", Wuhan University, Wuhan, June, 2023.
21. Invited talk on "Random Matrix Methods for Machine Learning: "Lossless" Compression of Large and Deep Neural Networks", NCIIP 2023, Changchun, Jinlin, May, 2023.
22. Invited talk on "Random Matrix Methods for Machine Learning: "Lossless" Compression of Large and Deep Neural Networks", SPMS MAS Seminar Series, School of Physical and Mathematical Sciences, Nanyang Technological University, Jan, 2023.
23. Invited talk on "Random Matrix Methods for Machine Learning: An Application to "Lossless" Compression of Large and Deep Neural Networks", SDS Workshop on "Topics in Random Matrix Theory", CUHK-Shenzhen, October, 2022.
24. Invited talk on "Random Matrix Methods for Machine Learning: "Lossless" Compression of Large Neural Networks", Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University, August, 2022.
25. Invited talk on "Random Matrix Methods for Machine Learning: "Lossless" Compression of Large Neural Networks", China conference on Scientific Machine Learning (CSML 2022), August, 2022.
26. Invited talk on "Random Matrix Methods for Machine Learning", Gaoling School of Artificial Intelligence, Renmin University of China, March, 2022.

27. Invited talk on "Performance-complexity Trade-off in Large Dimensional Spectral Clustering", HUST Vision and Learning Salon 2021, Huazhong University of Science and Technology, December, 2021.
28. Invited talk on "A Random Matrix Approach to Large Dimensional Machine Learning", AI+Math Colloquia, Institute of Natural Sciences, Shanghai Jiao Tong University, 2021.
29. Invited talk on "A Random Matrix Approach to Large Dimensional Machine Learning", STAT-DS Seminar, Department of Statistics and Data Science, Southern University of Science and Technology, 2021.
30. Invited talk on "A Random Matrix Approach to Large Dimensional Machine Learning", Optimization Seminar, Academy of Mathematics and Systems Science, Chinese Academy of Science, 2021.
31. Invited talk on "A Data-dependent Theory of Overparameterization: Phase Transition, Double Descent, and Beyond" at Workshop on the Theory of Over-parameterized Machine Learning (TOPML) 2021. April 20-21, 2021. Link to video.
32. Invited talk on "Performance-complexity Trade-off in Large Dimensional Spectral Clustering", Statistics Seminar, Research School of Finance, Actuarial Studies and Statistics, Australian National University, Canberra, 2021.
33. Invited talk on "Performance-complexity Trade-off in Large Dimensional Spectral Clustering", HUAWEI First Mini-workshop on Random Matrix Theory and Machine Learning, Paris, 2021.
34. Invited talk on "Performance-complexity Trade-off in Large Dimensional Spectral Clustering", STA 290 Seminar, Department of Statistics, University of California, Davis, 2021.
35. Invited talk on "Dynamical Aspects of Learning Linear Neural Networks", The Fields Institute for Research in Mathematical Sciences, *Second Symposium on Machine Learning and Dynamical Systems*, 2020. Link to recording.
36. Invited talk on "Random Matrix Advances in Large Dimensional Machine Learning", Shanghai University of Finance and Economics, *Random Matrices and Complex Data Analysis Workshop*, Shanghai, 2019.
37. Invited talk on "Random Matrix Viewpoint of Learning with Gradient Descent", DIMACS, *Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization*, Rutgers University, 2019. Link to recording.
38. Invited talk on "Recent Advances in Random Matrix Theory for Machine Learning and Neural Nets", workshop of the Matrix series on "*Random matrix theory faces information era*", Kraków, Poland, 2019. Link to recording.
39. Invited talk on "Dynamical Aspects of Deep Learning" (with Y. Chitour), *Séminaire d'Automatique du plateau de Saclay of iCODE institute*, Paris, France, 2019.
40. Invited talk on "Recent Advances in Random Matrix for Neural Networks", *Workshop on deep learning theory*, Shanghai JiaoTong University, China, 2018.
41. **Tutorial** on "Random Matrix Advances in Machine Learning and Neural Nets" (with R. Couillet and X. Mai), *The 26th European Signal Processing Conference (EUSIPCO'18)*, Roma, Italy, 2018.

References

- **Michael W. Mahoney**
  - Associate Adjunct Professor at Department of Statistics, UC Berkeley, CA, USA.
  - Director of the UC Berkeley FODA (Foundations of Data Analysis) Institute, Berkeley, CA, USA.
  - ✉ mmahoney@stat.berkeley.edu
- **Romain Couillet**
  - Full Professor at University Grenoble-Alps, France
  - Holder of the UGA MIAI LargeDATA Chair, University-Grenoble-Alps, France.
  - ✉ romain.couillet@gipsa-lab.grenoble-inp.fr
- **Yacine Chitour**
  - Full Professor at Pairs-Sud, University Paris-Saclay, France.
  - Director of the H-CODE institute, University Paris-Saclay, France.
  - ✉ yacine.chitour@l2s.centralesupelec.fr